Qingkai Shi

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SHORT BIO

Qingkai Shi is an associate professor in the School of Computer Science at Nanjing University. His research focuses on compiler techniques, especially formal program analysis, to rigorously ensure software quality. He has published extensively at premier venues in programming languages and software engineering. His research has received four ACM SIGPLAN or SIGSOFT Distinguished Paper Awards, one Google Research Paper Award, and the Hong Kong Ph.D. Fellowship.

Qingkai obtained his Ph.D. from the Hong Kong University of Science and Technology. He co-founded Sourcebrella LLC, where his research was commercialized. He then moved to Ant Group, which acquired Sourcebrella. Qingkai also enjoyed a wonderful period as a postdoctoral researcher at Purdue University.

EXPERIENCES

Nanjing University — 2023 -
- *Associate Professor*

Purdue University — 2021 - 2023
- *Postdoctoral Researcher*

Sourcebrella LLC, Ant Group — 2020 - 2021
- *Technical Expert*

Sourcebrella LLC — 2015 - 2020
- *Co-founder of Sourcebrella*

The Hong Kong University of Science and Technology — 2015 - 2020
- *Ph.D. in Computer Science*

PUBLICATIONS

SELECTED PUBLICATIONS

1. **Qingkai Shi**, Xiaoheng Xie, Xianjin Fu, Peng Di, Huawei Li, Ang Zhou, and Gang Fan. Datalog-Based Language-Agnostic Change Impact Analysis for Microservices. *Proceedings of the ACM/IEEE International Conference on Software Engineering (ICSE'25)*. 2025.
2. **Qingkai Shi**, Junyang Shao, Yapeng Ye, Mingwei Zheng, and Xiangyu Zhang. Lifting Network Protocol Implementation to Precise Format Specification with Security Applications. *Proceedings of the ACM Conference on Computer and Communications Security (CCS'23)*. 2023.
3. **Qingkai Shi**, Xiangzhe Xu, and Xiangyu Zhang. Extracting Protocol Format as State Machine via Controlled Static Loop Analysis. *Proceedings of the USENIX Security Symposium (SEC'23)*. 2023.

4. **Qingkai Shi**, Yongchao Wang, Peisen Yao, and Charles Zhang. Indexing the Extended Dyck-CFL Reachability for Context-Sensitive Program Analysis. *Proceedings of the ACM on Programming Languages (OOPSLA'22)*. 2022.

5. **Qingkai Shi**, Peisen Yao, Rongxin Wu, and Charles Zhang. Path-Sensitive Sparse Analysis without Path Conditions. *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI'21)*. 2021.

6. **Qingkai Shi** and Charles Zhang. Pipelining Bottom-up Data Flow Analysis. *Proceedings of the ACM/IEEE International Conference on Software Engineering (ICSE'20)*. 2020.

7. **Qingkai Shi**, Rongxin Wu, Gang Fan, and Charles Zhang. Conquering the Extensional Scalability Problem for Value-Flow Analysis Frameworks. *Proceedings of the ACM/IEEE International Conference on Software Engineering (ICSE'20)*. 2020.

8. **Qingkai Shi**, Xiao Xiao, Rongxin Wu, Jinguo Zhou, Gang Fan, and Charles Zhang. Pinpoint: Fast and Precise Sparse Value Flow Analysis for Million Lines of Code. *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI'18)*. 2018.

OTHER PUBLICATIONS

9. Yufan Zhou, Bing Shui, Hengdi Ye, Jielun Wu, and **Qingkai Shi**. Navigating the Labyrinth: Demystifying Vulnerabilities in Routing Protocol Implementations. *Proceedings of the Network and Distributed System Security Symposium (NDSS'27)*. 2027.

10. Hongcheng Fan, Jielun Wu, Xincheng He, Yang Feng, Yibiao Yang, Baowen Xu, and **Qingkai Shi**. A Survey of Symbol Name Recovery in Software Reverse Engineering. *Proceedings of the ACM Computing Surveys (CSUR'27)*. 2027.

11. Hengdi Ye, Bing Shui, Jielun Wu, Yufan Zhou, Baowen Xu, and **Qingkai Shi**. Generating Precise Format Specification for Network Protocols Through Adversarial LLM Interactions. *Proceedings of the USENIX Security Symposium (SEC'26)*. 2026.

12. Xinlong Wu, Ruiyu Zhou, Peisen Yao, and **Qingkai Shi**. Sound and Precise Symbolic Automata Model for Stateful Software Systems. *Proceedings of the International Conference on Computer Aided Verification (CAV'26)*. 2026.

13. Jiasheng Jiang, Mingwei Zheng, **Qingkai Shi**, and Xiangyu Zhang. SFA-Miner: Mining Path-Sensitive API Usage Patterns via Symbolic Finite Automata. *Proceedings of the IEEE Symposium on Security and Privacy (SP'26)*. 2026.

14. Chengpeng Wang, Yifei Gao, Wuqi Zhang, Xuwei Liu, Jinyao Guo, Mingwei Zheng, **Qingkai Shi**, and Xiangyu Zhang. NESA: Relational Neuro-Symbolic Static Program Analysis. *Proceedings of the ACM Symposium on the Foundations of Software Engineering (FSE'26)*. 2026.

15. Yuxuan He, Ruilin Jiang, He Zhang, **Qingkai Shi**, Huaxun Huang, and Rongxin Wu. Hermes: Making Path-Sensitive Pointer Analysis Scalable for Sparse Value-Flow Analysis. *Proceedings of the ACM on Programming Languages (OOPSLA'26)*. 2026.

16. Qingyang Li, Yibiao Yang, Maolin Sun, Jiangchang Wu, **Qingkai Shi**, and Yuming Zhou. Isolating Compiler Faults via Multiple Pairs of Adversarial Compilation Configurations. *ACM Transactions on Software Engineering and Methodology (TOSEM'26)*. 2026.

17. Qingyang Li, Yibiao Yang, Maolin Sun, Jiangchang Wu, **Qingkai Shi**, Yuming Zhou, and Baowen Xu. Boosting Compiler Fault Localization: Getting the Best of Both Worlds by Fusing Dynamic and Historical Data. *IEEE Transactions on Software Engineering (TSE'26)*. 2026.

18. Jielun Wu, Bing Shui, Hongcheng Fan, Shengxin Wu, Rongxin Wu, Yang Feng, Baowen Xu, and **Qingkai Shi**. Protecting Source Code Privacy When Hunting Memory Bugs. *Proceedings of the ACM/IEEE International Conference on Automated Software Engineering (ASE'25)*. 2025.

19. Bing Shui, Yufan Zhou, Jielun Wu, Baowen Xu, and **Qingkai Shi**. Validating Interior Gateway Routing Protocols via Equivalent Topology Synthesis. *Proceedings of the ACM Conference on Computer and Communications Security (CCS'25)*. 2025.

20. Zixi Liu, Yang Feng, Yunbo Ni, Shaohua Li, Xizhe Yin, **Qingkai Shi**, Baowen Xu, and Zhendong Su. An Empirical Study of Bugs in the Rustc Compiler. *Proceedings of the ACM on Programming Languages (OOPSLA'25)*. 2025.

21. Mingwei Zheng, Danning Xie, **Qingkai Shi**, Chengpeng Wang, and Xiangyu Zhang. Validating Network Protocol Parsers with Traceable RFC Document Interpretation. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'25)*. 2025.

22. Mingwei Zheng, **Qingkai Shi**, Xuwei Liu, Xiangzhe Xu, Le Yu, Congyu Liu, Guannan Wei, and Xiangyu Zhang. ParDiff: Practical Static Differential Analysis of Network Protocol Parsers. *Proceedings of the ACM on Programming Languages (OOPSLA'24)*. 2024. **ACM SIGPLAN Distinguished Paper Award**

23. Peisen Yao, Jinguo Zhou, Xiao Xiao, **Qingkai Shi**, Rongxin Wu, and Charles Zhang. Falcon: A Fused Approach to Path-Sensitive Sparse Data Dependence Analysis. *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI'24)*. 2024.

24. Wuqi Zhang, Zhuo Zhang, **Qingkai Shi**, Lu Liu, Lili Wei, Yepang Liu, Xiangyu Zhang, and Shing-Chi Cheung. Nyx: Detecting Exploitable Front-Running Vulnerabilities in Smart Contracts. *Proceedings of the IEEE Symposium on Security and Privacy (SP'24)*. 2024.

25. Shiwei Feng, Yapeng Ye, **Qingkai Shi**, Zhiyuan Cheng, Xiangzhe Xu, Siyuan Cheng, Hongjun Choi, and Xiangyu Zhang. ROCAS: Root Cause Analysis of Autonomous Driving Accidents via Cyber-Physical Co-mutation. *Proceedings of the ACM/IEEE International Conference on Automated Software Engineering (ASE'24)*. 2024. **ACM SIGSOFT Distinguished Paper Award**

26. Rongxin Wu, Y. He, J. Huang, C. Wang, W. Tang, **Qingkai Shi**, X. Xiao, and Charles Zhang. A Two-Layer Persistent Summary Design for Taming Third-Party Libraries in Static Bug-Finding Systems. *Proceedings of the ACM/IEEE International Conference on Software Engineering (ICSE'24)*. 2024.

27. Xizhe Yin, Yang Feng, **Qingkai Shi**, Zixi Liu, Hongwang Liu, and Baowen Xu. FRIES: Fuzzing Rust Library Interactions via Efficient Ecosystem-Guided Target Generation. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'24)*. 2024.

28. Yi Sun, Chengpeng Wang, Gang Fan, **Qingkai Shi**, and Xiangyu Zhang. Fast and Precise Static Null Exception Analysis with Synergistic Preprocessing. *IEEE Transactions on Software Engineering (TSE'24)*. 2024.

29. Xinmeng Xia, Yang Feng, **Qingkai Shi**, James Jones, Xiangyu Zhang, and Baowen Xu. Enumerating Valid Non-Alpha-Equivalent Programs for Interpreter Testing. *ACM Transactions on Software Engineering and Methodology (TOSEM'24)*. 2024.

30. Yapeng Ye, Zhuo Zhang, **Qingkai Shi**, Yousra Aafer, and Xiangyu Zhang. D-ARM: Disassembling ARM Binaries by Lightweight Superset Instruction Interpretation and Graph Modeling. *Proceedings of the IEEE Symposium on Security and Privacy (SP'23)*. 2023.

31. Xiangzhe Xu, Z. Xun, S. Feng, S. Chen, Y. Ye, **Qingkai Shi**, G. Tao, L. Yu, Z. Zhang, and Xiangyu Zhang. PEM: Representing Binary Program Semantics for Similarity Analysis via A Probabilistic Execution Model. *Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'23)*. 2023.

32. Xiangzhe Xu, S. Feng, Y. Ye, G. Shen, Z. Su, S. Chen, G. Tao, **Qingkai Shi**, Z. Zhang, and Xiangyu Zhang. Improving Binary Code Similarity Transformer Models by Semantics-driven Instruction Deemphasis. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'23)*. 2023.

33. Heqing Huang, Hung-Chun Chiu, **Qingkai Shi**, Peisen Yao, and Charles Zhang. Balancing Seed Scheduling via Monte Carlo Planning. *IEEE Transactions on Dependable and Secure Computing (TDSC'23)*. 2023.

34. Chengpeng Wang, Wenyang Wang, Peisen Yao, **Qingkai Shi**, Jinguo Zhou, Xiao Xiao, and Charles Zhang. Anchor: Fast and Precise Value-Flow Analysis for Containers via Memory Orientation. *ACM Transactions on Software Engineering and Methodology (TOSEM'23)*. 2023.

35. Chengpeng Wang, Peisen Yao, Wensheng Tang, **Qingkai Shi**, and Charles Zhang. Complexity-Guided Container Replacement Synthesis. *Proceedings of the ACM on Programming Languages (OOPSLA'22)*. 2022. **ACM SIGPLAN Distinguished Paper Award**

36. Yuandao Cai, Chengfeng Ye, **Qingkai Shi**, and Charles Zhang. Peahen: Fast and Precise Static Deadlock Detection via Context Reduction. *Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'22)*. 2022.

37. Yiyuan Guo, Jinguo Zhou, Peisen Yao, **Qingkai Shi**, and Charles Zhang. Precise Divide-By-Zero Detection with Affirmative Evidence. *Proceedings of the ACM/IEEE International Conference on Software Engineering (ICSE'22)*. 2022.

38. Heqing Huang, Yiyuan Guo, **Qingkai Shi**, Peisen Yao, Rongxin Wu, and Charles

Zhang. Beacon: Directed Grey-Box Fuzzing with Provable Path Pruning. *Proceedings of the IEEE Symposium on Security and Privacy (SP'22)*. 2022. **Google Research Paper Reward**

39. Peisen Yao, **Qingkai Shi**, Heqing Huang, and Charles Zhang. Program Analysis via Efficient Symbolic Abstraction. *Proceedings of the ACM on Programming Languages (OOPSLA'21)*. 2021.

40. Peisen Yao, Heqing Huang, Wensheng Tang, **Qingkai Shi**, Rongxin Wu, and Charles Zhang. Skeletal Approximation Enumeration for SMT Solver Testing. *Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'21)*. 2021.

41. Peisen Yao, Heqing Huang, Wensheng Tang, **Qingkai Shi**, Rongxin Wu, and Charles Zhang. Fuzzing SMT Solvers via Two-Dimensional Input Space Exploration. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'21)*. 2021.

42. Heqing Huang, Peisen Yao, Rongxin Wu, **Qingkai Shi**, and Charles Zhang. Pangolin: Incremental Hybrid Fuzzing with Polyhedral Path Abstraction. *Proceedings of the IEEE Symposium on Security and Privacy (SP'20)*. 2020.

43. Peisen Yao, **Qingkai Shi**, Heqing Huang, and Charles Zhang. Fast Bit-Vector Satisfiability. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'20)*. 2020.

44. Gang Fan, Chengpeng Wang, Rongxin Wu, Xiao Xiao, **Qingkai Shi**, and Charles Zhang. Escaping Dependency Hell: Finding Build Dependency Errors with the Unified Dependency Graph. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'20)*. 2020.

45. Yang Feng, **Qingkai Shi**, Xinyu Gao, Jun Wan, Chunrong Fang, and Zhenyu Chen. DeepGini: Prioritizing Massive Tests to Enhance the Robustness of Deep Neural Networks. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'20)*. 2020.

46. Chunrong Fang, Zixi Liu, Yangyang Shi, Jeff Huang, and **Qingkai Shi**. Functional Code Clone Detection with Syntax and Semantics Fusion Learning. *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'20)*. 2020.

47. Gang Fan, Rongxin Wu, **Qingkai Shi**, Xiao Xiao, Jinguo Zhou, and Charles Zhang. Smoke: Scalable Path-Sensitive Memory Leak Detection for Millions of Lines of Code. *Proceedings of the ACM/IEEE International Conference on Software Engineering (ICSE'19)*. 2019. **ACM SIGSOFT Distinguished Paper Award**

48. **Qingkai Shi**, Jeff Huang, Zhenyu Chen, and Baowen Xu. Verifying Synchronization for Atomicity Violation Fixing. *IEEE Transactions on Software Engineering (TSE'16)*. 2016.

49. **Qingkai Shi**, Zhenyu Chen, Chunrong Fang, Yang Feng, and Baowen Xu. Measuring the Diversity of a Test Set with Distance Entropy. *IEEE Transactions on Reliability (TR'16)*. 2016.

PATENTS

Defect detection method, device, system, and computer readable medium.

- US Patent No. 20190108003
- China Patent No. 201811013103, 201811013000, 201811015751, 2018110146864

SERVICE

Journal Reviewer

- *ACM Computing Surveys*
- *ACM Transactions on Software Engineering and Methodology*
- *IEEE Transactions on Software Engineering*
- *IEEE Transactions on Reliability*
- *IEEE Transactions on Emerging Topics in Computing*
- *Springer Automated Software Engineering*

Member of Program Committee

- *ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI'27)*
- *ACM International Conference on the Foundations of Software Engineering (FSE'27)*
- *ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA'26)*
- *ACM/IEEE International Conference on Automated Software Engineering (ASE'24, ASE'25, ASE'26)*
- *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'25)*
- *International Conference on Internetware (Internetware'26)*
- *International Symposium on Dependable Software Engineering: Theories, Tools and Applications (SETTA'26)*

Member of External Review Committee

- *ACM International Conference on the Foundations of Software Engineering (FSE'25)*
- *European Conference on Object-Oriented Programming (ECOOP'23)*

Member of Artifact Evaluation Committee

- *ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA'24)*
- *ACM Symposium on Principles of Programming Languages (POPL'22, POPL'23)*
- *European Conference on Object-Oriented Programming (ECOOP'23)*

Member of Organizing Committee

- *ACM SIGPLAN International Workshop on Language Models and Programming Languages @ SPLASH'25, SPLASH'26*
- *Symposium on Novel Technology for Software Analysis and Security @ ChinaSoft'25, ChinaSoft'26*

TEACHING

Data Structure (Spring 2024, 2025; Fall 2025)

Interpreters and Virtual Machines (Spring 2025, 2026)

Compilers (Spring 2024, 2025)